

MySQL Date & Time Complete Cheat Sheet

1. Current Date and Time

```
SELECT CURDATE();      -- Current Date
SELECT CURRENT_DATE(); -- Current Date

SELECT CURTIME();      -- Current Time
SELECT CURRENT_TIME(); -- Current Time

SELECT NOW();          -- Current Date & Time
SELECT CURRENT_TIMESTAMP();
SELECT SYSDATE();      -- System Date & Time
```

Example Output

```
CURDATE()    : 2026-05-29
CURTIME()    : 15:30:45
NOW()        : 2026-05-29 15:30:45
```

2. Extract Parts from Date

```
SELECT YEAR('2026-05-29');
SELECT MONTH('2026-05-29');
SELECT DAY('2026-05-29');

SELECT QUARTER('2026-05-29');
SELECT WEEK('2026-05-29');
SELECT WEEKDAY('2026-05-29');

SELECT DAYNAME('2026-05-29');
SELECT MONTHNAME('2026-05-29');

SELECT DAYOFMONTH('2026-05-29');
SELECT DAYOFYEAR('2026-05-29');
SELECT DAYOFWEEK('2026-05-29');
```

3. Extract Parts from DateTime

```
SELECT HOUR('2026-05-29 15:30:45');
SELECT MINUTE('2026-05-29 15:30:45');
SELECT SECOND('2026-05-29 15:30:45');

SELECT MICROSECOND('2026-05-29 15:30:45.123456');
```

4. DATE_FORMAT()

Common Formats

```
SELECT DATE_FORMAT(NOW(), '%d-%m-%Y');
SELECT DATE_FORMAT(NOW(), '%d/%m/%Y');
SELECT DATE_FORMAT(NOW(), '%M %d, %Y');
SELECT DATE_FORMAT(NOW(), '%W');
SELECT DATE_FORMAT(NOW(), '%h:%i:%s %p');
```

Format Specifiers

Format	Meaning
%Y	4 digit year
%y	2 digit year
%m	Month number
%M	Month name
%d	Day
%W	Weekday name
%H	Hour (24)
%h	Hour (12)
%i	Minutes
%s	Seconds

%p AM/PM

5. Date Arithmetic

Add Days

```
SELECT DATE_ADD(CURDATE(), INTERVAL 5 DAY);
```

Add Months

```
SELECT DATE_ADD(CURDATE(), INTERVAL 2 MONTH);
```

Add Years

```
SELECT DATE_ADD(CURDATE(), INTERVAL 1 YEAR);
```

Add Hours

```
SELECT DATE_ADD(NOW(), INTERVAL 5 HOUR);
```

6. DATE_SUB()

```
SELECT DATE_SUB(CURDATE(), INTERVAL 10 DAY);
```

```
SELECT DATE_SUB(CURDATE(), INTERVAL 2 MONTH);
```

```
SELECT DATE_SUB(CURDATE(), INTERVAL 1 YEAR);
```

7. DATEDIFF()

Difference in days.

```
SELECT DATEDIFF('2026-12-31','2026-05-29');
```

Output:

8. TIMESTAMPDIFF()

Difference in Years

```
SELECT TIMESTAMPDIFF(  
    YEAR,  
    '2015-01-01',  
    CURDATE()  
);
```

Difference in Months

```
SELECT TIMESTAMPDIFF(  
    MONTH,  
    '2025-01-01',  
    CURDATE()  
);
```

Difference in Days

```
SELECT TIMESTAMPDIFF(  
    DAY,  
    '2025-01-01',  
    CURDATE()  
);
```

Difference in Hours

```
SELECT TIMESTAMPDIFF(  
    HOUR,  
    '2026-05-28 10:00:00',  
    NOW()  
);
```

9. LAST_DAY()

```
SELECT LAST_DAY('2026-05-15');
```

Output

2026-05-31

10. MAKEDATE()

```
SELECT MAKEDATE(2026,150);
```

Output

2026-05-30

11. STR_TO_DATE()

Convert String to Date

```
SELECT STR_TO_DATE(  
    '29-05-2026',  
    '%d-%m-%Y'  
);
```

12. DATE()

Extract Date Only

```
SELECT DATE(NOW());
```

Output

2026-05-29

13. TIME()

Extract Time Only

```
SELECT TIME(NOW());
```

Output

15:30:45

14. UNIX_TIMESTAMP()

```
SELECT UNIX_TIMESTAMP();
```

```
SELECT UNIX_TIMESTAMP(  
    '2026-05-29 15:30:45'  
);
```

15. FROM_UNIXTIME()

```
SELECT FROM_UNIXTIME(  
    1780050000  
);
```

16. Time Difference

```
SELECT TIMEDIFF(  
    '18:00:00',  
    '09:30:00'  
);
```

Output

08:30:00

17. Add Time

```
SELECT ADDTIME(  
    '10:00:00',  
    '02:30:00'  
);
```

Output

12:30:00

18. Subtract Time

```
SELECT SUBTIME(  
    '10:00:00',  
    '02:00:00'  
);
```

Output

08:00:00

19. Common WHERE Clause Date Filters

Today's Records

```
SELECT *  
FROM orders  
WHERE DATE(order_date)=CURDATE();
```

Yesterday's Records

```
SELECT *  
FROM orders  
WHERE DATE(order_date)=CURDATE()-INTERVAL 1 DAY;
```

Last 7 Days

```
SELECT *  
FROM orders  
WHERE order_date >= CURDATE()-INTERVAL 7 DAY;
```

Current Month

```
SELECT *
```

```
FROM orders
WHERE MONTH(order_date)=MONTH(CURDATE())
AND YEAR(order_date)=YEAR(CURDATE());
```

Current Year

```
SELECT *
FROM orders
WHERE YEAR(order_date)=YEAR(CURDATE());
```

Between Two Dates

```
SELECT *
FROM orders
WHERE order_date BETWEEN
    '2026-01-01'
    AND
    '2026-12-31';
```

20. Age Calculation

```
SELECT emp_name,
    TIMESTAMPDIFF(
        YEAR,
        dob,
        CURDATE()
    ) AS age
FROM employees;
```

21. Experience Calculation

```
SELECT emp_name,
    TIMESTAMPDIFF(
        YEAR,
        hire_date,
        CURDATE()
    ) AS experience
FROM employees;
```

22. First Day of Current Month

```
SELECT DATE_FORMAT(  
    CURDATE(),  
    '%Y-%m-01'  
);
```

23. Last Day of Current Month

```
SELECT LAST_DAY(CURDATE());
```

24. First Day of Current Year

```
SELECT MAKEDATE(  
    YEAR(CURDATE()),  
    1  
);
```

25. Real Interview Scenarios

Employees Joined in Last 30 Days

```
SELECT *  
FROM employees  
WHERE hire_date >=  
    CURDATE() - INTERVAL 30 DAY;
```

Orders Placed This Week

```
SELECT *  
FROM orders  
WHERE YEARWEEK(order_date)  
    = YEARWEEK(CURDATE());
```

Birthday This Month

```
SELECT *  
FROM employees  
WHERE MONTH(dob)  
      = MONTH(CURDATE());
```

Records Older Than 1 Year

```
SELECT *  
FROM orders  
WHERE order_date <  
      CURDATE() - INTERVAL 1 YEAR;
```

Latest Record

```
SELECT *  
FROM orders  
ORDER BY order_date DESC  
LIMIT 1;
```

Latest Order Per Customer

```
SELECT customer_id,  
      MAX(order_date) AS latest_order  
FROM orders  
GROUP BY customer_id;
```

Most Frequently Used in Projects

```
NOW()  
CURDATE()  
CURTIME()
```

```
DATE()  
TIME()
```

```
DATE_ADD()  
DATE_SUB()
```

```
DATEDIFF()  
TIMESTAMPDIFF()
```

```
YEAR()  
MONTH()  
DAY()
```

DATE_FORMAT()

STR_TO_DATE()

LAST_DAY()

UNIX_TIMESTAMP()

FROM_UNIXTIME()

These functions cover about **95% of real-world Data Engineering, ETL, Reporting, BI, Data Warehouse, MySQL, PostgreSQL, Oracle, SQL Server, and Spark SQL date/time requirements.**